

Supply Chain Resiliency & Margin Protection in Prestige Beauty

An Enterprise Integration Case Study: Final Report

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1. Introduction

Prestige beauty and luxury consumer packaged goods brands operate on unusually high gross margins, typically between 70% and 80%, protected historically by strong brand equity, proprietary formulations, and tightly controlled distribution networks. That protection is now under pressure. Cross-border trade tariffs, geopolitical friction along major shipping lanes, and stricter chemical-traceability compliance frameworks have exposed a structural vulnerability shared across the industry: an over-reliance on concentrated, single-sourced production and packaging nodes. When a single supplier node is disrupted, the result is an immediate retail stockout across high-velocity SKUs, with no secondary routing path available to defend revenue.

This project asks a specific, decision-relevant question: **how can a prestige beauty brand transition from a vulnerable, single-factory sourcing matrix to a resilient, multi-supplier network without diluting its retail gross margins?** Rather than treating supply chain risk and margin protection as two separate analyses, this project builds a single integrated scenario-modeling framework that connects supplier quality risk, tariff-driven cost exposure, and consumer demand signals into one decision picture.

What distinguishes this analysis from a conventional supply chain EDA is the enterprise integration lens. Rather than stopping at descriptive statistics, each finding in this report is explicitly mapped to where it would land in a real enterprise stack: a margin-compression finding is framed as an SAP S/4HANA or Oracle Cloud ERP costing-view trigger; a stockout-risk finding is framed as a Manhattan Active WMS reorder-point recalibration; and a review-velocity finding is framed as a demand-sensing input to Anaplan or o9 Solutions. This framing is what gives the analysis its value as an enterprise proof-of-concept.

A scope note carried through every deliverable in this project: the primary operational dataset contains 100 SKUs. The analysis is therefore intentionally directional. It demonstrates a methodology and a relative risk ranking across product types, suppliers, and transportation modes, rather than producing a statistically generalizable forecast. This limitation is disclosed transparently throughout this report rather than glossed over, consistent with the project's broader commitment to honest, defensible analysis.

2. Business Impact

Findings from this project are designed to inform three enterprise decision layers, each mapped to a specific system an organization would already be operating.

Sourcing Strategy

Identifying which suppliers and product categories carry the highest defect concentration and lead-time instability enables data-driven vendor diversification decisions. The variance between baseline and tariff-adjusted margins can be mapped directly to an ERP costing view (SAP S/4HANA or Oracle Cloud ERP), triggering an automated sourcing review whenever margin crosses below a corporate-mandated floor.

Margin Protection

Modeling the tariff_impact_coefficient across product categories gives an S&OP team a quantified view of COGS inflation risk under 2026 trade conditions, rather than a qualitative sense that "tariffs are a problem." This is the analytical layer a CFO would use to decide which category to move first.

Demand Sensing

Correlating supplier lead-time variance against consumer demand metrics establishes a model for dynamically recalculating Safety Stock and Reorder Points inside a WMS platform such as Manhattan Active, mitigating stockout risk for high-velocity products. Integrating consumer-facing signals such as review velocity into upstream planning tools like Anaplan or o9 Solutions shifts a supply chain from a reactive posture to a predictive one.

Taken together, these three layers are why this project is framed as an enterprise integration case study: each finding is only as useful as the system it can be acted on inside, and this project deliberately names that system for every finding it produces.

3. Data

Three data layers were integrated to build this analysis. Each is documented in full in the project's Data Curation Report; this section summarizes the sourcing and structure of each.

Dataset 1: Supply Chain Analysis (Primary Operational Data)

Sourced from Kaggle (Singh, 2023), this dataset captures end-to-end procurement, logistics, and manufacturing metrics for 100 SKUs across skincare (40%), haircare (34%), and cosmetics (26%). It spans 24 original columns with zero null values, covering price, stock levels, order quantities, supplier and delivery lead times, manufacturing costs, defect rates, inspection results, shipping carriers, transportation modes, and routes.

Dataset 2: Sephora Products and Customer Reviews (Secondary Demand Signal)

Sourced from Kaggle (Inky, 2023), this dataset provided a product catalog (8,494 products, 27 columns) and a reviews corpus (1,094,411 rows across five files, spanning August 2008 to March 2023), scraped from Sephora.com and covering prestige brands including Dior, Lancôme, Clinique, and TOM FORD.

A critical data-scope finding surfaced during the Tableau build phase, after the EDA had already been submitted: every product_id in the reviews files maps exclusively to the Skincare category in product_info's primary_category field. There is zero overlap with Makeup or Hair. This means the reviews dataset is, in practice, a skincare-only demand signal, despite the broader "Sephora Products and Skincare Reviews" naming suggesting wider coverage. This is a genuine property of the underlying Kaggle data, not a defect introduced by this project's cleaning process. Its implications are discussed further in Section 7 (Challenges) and disclosed directly on the datafolio and dashboard wherever review-based findings appear.

Dataset 3: External Trade Policy Benchmarks

USTR Section 301 tariff schedules and EU Commission Chapter 33 customs tariff rates were referenced directly (no file download required) to validate the tariff percentages assigned to each product category, grounding the project's core cost model in real, cited trade policy rather than an arbitrary assumption.

4. Data Cleaning and Wrangling

Dataset 1 arrived with zero null values across all 24 columns, so no null-imputation was required. Cleaning instead focused on renaming ambiguous fields for enterprise clarity and on engineering new analytical fields. "Lead times" and "Lead time" were renamed to `supplier_lead_time` and `delivery_lead_time` respectively, since the two represented distinct planning stages that had been easy to confuse under their original names:

- **tariff_impact_coefficient:** an engineered proxy variable mapping each `product_type` to a 2026 trade tariff tier: cosmetics 25% (Section 301 East Asia packaging/componentry exposure), skincare 10% (EU MFN premium on active-ingredient and luxury glass imports), haircare 0% (domestic/near-shore baseline). This Structural Attribute Proxy Strategy was necessary because the dataset's geographic metadata is limited to five anonymized domestic hubs, preventing a direct join to macro-level trade data.
- **tariff_adjusted_mfg_cost:** $\text{manufacturing_costs} \times (1 + \text{tariff_impact_coefficient})$, the core cost variable driving every margin-compression finding in this report.
- **lead_time_variance:** the absolute difference between `supplier_lead_time` and `delivery_lead_time`, quantifying planning-buffer risk per SKU.
- **stock_to_sales_ratio:** $\text{stock_levels} \div \text{number_of_products_sold}$, the inventory-health proxy used to identify stockout risk.

For Dataset 2, cleaning focused on dropping fields with extreme null rates. `value_price_usd` and `sale_price_usd` were both over 94% null and were removed, along with `helpfulness` (55.1% null) and several catalog fields exceeding 67% null. `review_year` and `review_month` were engineered from `submission_time` to support the time-series demand-velocity analysis in Section 5.

Bridging the Datasets

Datasets 1 and 2 do not share a direct key. Dataset 1 uses generic SKU0 to SKU99 identifiers, while Dataset 2 uses Sephora's own `product_id`. A structural category-bridge mapping (`product_type` → `primary_category`: skincare→Skincare, cosmetics→Makeup, haircare→Hair) was used to connect the two at the category level. This bridge was validated in EDA Analysis #9 and found analytically sound: skincare shows both the highest Sephora rating (4.31) and the highest demand pressure in the operational data (lowest stock-to-sales ratio, 0.13), a convergence that would not be expected if the bridge mapping were spurious. A separate `category_bridge.csv` file was also connected during the Tableau build to support this same join at the visualization layer, though it was ultimately not needed once Tab 3's review chart was simplified to skincare-only. See Section 7 for the full scope-decision record.

5. Analysis Methods

Nine analyses were conducted across the two datasets; the five most decision-relevant are summarized here, each shown with the finding it produced and the enterprise action it implies.

Lead Time Variance and Defect Concentration

Lead time variance between supplier_lead_time and delivery_lead_time is right-skewed, averaging 9.9 days with a maximum of 28 days; 15 SKUs exceed 20 days of variance and represent the highest-priority candidates for near-shore dual-sourcing. Defect rate by product type initially contradicted the project's working hypothesis: haircare carries the highest average defect rate (2.48%), ahead of skincare (2.33%) and cosmetics (1.92%), despite haircare bearing zero tariff exposure. This is a genuinely important finding, not a null result. It demonstrates that **quality risk and tariff risk are independent dimensions** requiring separate mitigation strategies, since a category can be low-risk on one axis and high-risk on the other.

Supplier-Level Quality Concentration

Cross-tabulating inspection results against the five anonymized suppliers surfaced the project's single most actionable finding: **Supplier 4 carries a 66.7% inspection failure rate**, with 12 failed inspections and zero passed inspections across 18 SKUs, making it the clear, single-source quality bottleneck in the portfolio. Supplier 2 (36.4%) and Supplier 5 (38.9%) also exceed the portfolio average failure rate of 36%. This is precisely the sourcing concentration risk the project's dual-sourcing thesis is built to address, and it became the datafolio's headline statistic.

Tariff-Adjusted Margin Impact

Applying the tariff_impact_coefficient to manufacturing costs produced a directionally clear, if numerically modest, result: cosmetics margin compresses by 0.21 percentage points and skincare by 0.13 points, while haircare is unaffected. The absolute compression is small only because this sample's baseline margins run near 98%, far above the 70–80% margins typical of real enterprise beauty portfolios. At real-world margin levels, the same percentage-point ranking (cosmetics > skincare > haircare) would translate into materially larger COGS inflation. The ranking, not the absolute magnitude, is the analytically significant output of this model, and this distinction is disclosed explicitly rather than left for the reader to assume.

Stockout Risk and Consumer Demand Signal

Plotting stock levels against units sold identified 26 SKUs across all categories with a stock-to-sales ratio below 0.05. These are critical stockout risk zones. Skincare alone accounts for 14 of those 26 SKUs and carries the lowest category-level stock-to-sales ratio (0.13) and the highest average units sold (518), making it the most demand-pressured category in the portfolio. This finding is reinforced independently by the (skincare-only) Sephora review data: review volume grew 6.5x between 2015 and 2020, a clear leading indicator that, read together with the stockout finding, supports treating review-velocity monitoring as a legitimate advance-warning signal for demand planning. This is precisely the S&OP integration use case named in this project's original scope.

Transportation Mode Trade-offs

Comparing shipping cost and delivery lead time across transportation modes produced a counterintuitive result: Air is the most expensive mode (avg \$6.02/unit) and also carries the longest average supplier lead time (18.27 days), while Sea is both the cheapest (avg \$4.97/unit) and fastest (12.18 days). This suggests the dataset's transportation_modes field reflects the outbound distribution leg rather than the full supply chain, and that this portfolio's Sea routes are optimized short-haul lanes rather than long transoceanic ones. This is a useful caveat for anyone reusing this dataset's transport-mode field at face value.

Statistical Analysis & Model Validity

No predictive statistical model (regression, classification, or time-series forecasting) was built for this project, and that omission was a deliberate scoping decision rather than an oversight. With only 100 SKUs and a sample whose baseline economics (98%+ gross margin) diverge meaningfully from real-world prestige beauty margins (70–80%), a predictive model trained on this data would carry a false sense of statistical rigor without a correspondingly valid claim to generalizability.

Instead, the project's core analytical instrument is the **tariff_impact_coefficient**, a deterministic, rules-based scenario model rather than a fitted statistical one. Its validity rests on a different basis than a predictive model's: it is not evaluated by out-of-sample accuracy, but by whether its category-tier assignments are defensibly grounded in real, cited trade policy (USTR Section 301 schedules and EU Commission Chapter 33 tariff rates), and by whether the ranking it produces (cosmetics > skincare > haircare in tariff exposure) is directionally consistent with the underlying policy basis. On both counts, the model holds. Where the model is explicitly not valid is in producing an absolute dollar-magnitude forecast for any real enterprise's COGS inflation. That would require the model to be recalibrated against that enterprise's actual margin structure, SKU count, and sourcing footprint, which is named directly as future work in Section 9.

A second, unbuilt parameter, Tariff Sensitivity (a user-adjustable 0 to 40% slider), was constructed during the Tableau build phase to support dynamic what-if scenario modeling, but was ultimately descope from the delivered dashboard in favor of the fixed, category-based **tariff_impact_coefficient**. This decision, along with several other deliberate scope decisions made across the project, is discussed in full in Section 7.

6. Dashboard

The dashboard is a three-tab, interactive Tableau Public workbook, explanatory (not exploratory) in design: each tab is deliberately scoped to a specific enterprise stakeholder identified in the Project Scoping document, surfacing only the KPIs, filters, and visual encodings relevant to that stakeholder's decision-making context, rather than presenting one generic view for all audiences.

<https://public.tableau.com/app/profile/shari.nishida/viz/CorrelationOneDataAnalyticsCapstone/ExecutiveSOPRiskMatrix?publish=yes>

Tab 1: Executive S&OP Risk Matrix (Supply Chain Director view)

Surfaces four portfolio-wide KPIs (Avg Sephora Rating, Avg Defect Rate, Avg Lead Time, Tariff-Adjusted COGS Inflation) alongside Defect Rate by Product Type, Lead Time Variability by Supplier, and a Cost vs. Revenue by SKU scatter under the tariff scenario overlay. Product Type, Supplier, and Transportation Modes filters let a Supply Chain Director isolate risk by category or vendor while the KPI tiles remain fixed at portfolio-wide totals, preserving a consistent benchmark regardless of what the charts are filtered to.

Tab 2: Margin Compression & Simulation (CFO view)

Quantifies the **tariff_impact_coefficient**'s directional gross-margin impact via a Margin Waterfall chart and a Baseline vs. Adjusted Margin comparison. A Near-Shore Volume Shift % parameter drives a live Simulated Blended Cost calculation, letting a CFO model the cost effect of shifting procurement volume toward near-shore suppliers in real time. As on Tab 1, filters apply only to the two charts, leaving the four KPI tiles as stable, portfolio-wide reference points.

Tab 3: Omnichannel Inventory & WMS Calibration (Warehouse Operations view)

Combines a skincare-scoped Review Volume vs. Stock Posture chart (explicitly captioned as skincare-only, per Section 3), a Reorder Point Table ranking the top SKUs by stockout risk, and a Stockout Risk Scatter with a Critical Stockout Zone reference line at Stock Level = 20, matching the threshold established in the EDA. A Product Type filter lets Warehouse Operations isolate reorder priorities and stockout risk by category.

Use Case

A stakeholder opens the dashboard and selects the tab matching their role. Within a tab, they filter by Product Type, Supplier, or Transportation Mode to narrow the view to their specific concern. For example, a CFO might isolate skincare's margin compression on Tab 2, or a Warehouse Operations lead might isolate cosmetics stockout risk on Tab 3, while portfolio-wide KPI benchmarks remain visible throughout for comparison. The three tabs are designed to be read as one connected narrative rather than three unrelated views: Tab 1 identifies where sourcing risk concentrates, Tab 2 quantifies what that risk costs in margin terms, and Tab 3 translates both findings into an actionable, SKU-level reorder-priority view a Warehouse Operations team could act on directly.

7. Challenges

The Sephora reviews dataset is skincare-only, not portfolio-wide

This was the most significant challenge encountered, because it surfaced after the EDA had already been submitted, during the Tableau dashboard build. A direct product_id cross-match against product_info's primary_category confirmed zero overlap into Makeup or Hair. Every review in the corpus belongs to a skincare product. The solution was disclosure, not concealment: rather than quietly reframing or hiding the affected findings (the datafolio's review-growth-by-price-tier statistic and Tab 3's review-velocity chart), both were explicitly captioned as skincare-scoped on the datafolio, on the dashboard chart title itself, and in this report. Tab 3's review chart was also rebuilt from its originally planned three-category form down to a skincare-only form once the limitation was confirmed, rather than left in a form that implied broader coverage than the data actually supports.

Scope Decisions: What Was Deliberately Left on the Sidelines

Several elements named or implied in the earlier Project Scoping document were consciously not built, or built and then not shipped, over the course of this project. Each is listed here explicitly, with the reasoning behind the decision, so that a gap between the original scope and the final deliverable reads as a documented judgment call rather than an oversight.

- **Tariff Sensitivity slider (0–40%).** Built as a parameter and tested end-to-end, but never wired into any dashboard chart. Descoped in favor of the fixed, category-based tariff_impact_coefficient, which is grounded in cited USTR/EU trade policy rather than an arbitrary user-adjustable value. Noted in the final Dashboard Submission as an intentional design decision.
- **"Margin recovery velocity" metric.** Named in the original Project Scoping document's prose as a core Tab 2 metric, but no formula or prior computation for it was ever found across three targeted searches of the project's own past work. Rather than invent a metric retroactively to match the scope document, it was dropped from the delivered dashboard.

- **"70% primary / 30% near-shore" volume split.** Referenced in the Project Scoping document as a worked example of a dual-sourcing scenario. Confirmed to be illustrative prose only, not a literal computed scenario. The dashboard's actual Near-Shore Volume Shift % parameter allows a user to test any split directly, making a single hardcoded 70/30 example unnecessary.
- **category_bridge.csv.** Connected as a data source during the Tableau build to support a bridged join between the operational and Sephora datasets on Tab 3. Ultimately not used, once Tab 3's review chart was simplified to skincare-only following the discovery that the reviews dataset carries no cosmetics or haircare coverage (Section 3).
- **Tab 1's Tariff Impact Coefficient filter.** Considered for the dashboard's filter panel, but deferred as redundant, since tariff_impact_coefficient maps to product_type on a strict one-to-one basis. A separate filter on it would duplicate, rather than add to, the existing Product Type filter.
- **Tab 2 waterfall bookend bars.** An explicit "Base" and "Adjusted" total bar bookending the Margin Waterfall chart was considered as a nice-to-have visual anchor, but not built, given time constraints relative to its marginal value on top of the chart's existing per-category bars.

8. Conclusion

Key Findings

1. **Supplier concentration, not tariffs, is the primary margin threat.** Supplier 4's 66.7% inspection failure rate is the single largest identified risk in the portfolio and should be the first target for dual-sourcing substitution, ahead of any tariff-driven mitigation.
2. **Quality risk and tariff risk are independent dimensions.** Haircare carries zero tariff exposure yet the highest defect rate (2.48%) in the portfolio, contradicting the project's initial hypothesis and confirming that these two risk types require separate mitigation strategies rather than a single combined one.
3. **Skincare is the highest-priority category for resilience investment.** It converges on every risk dimension measured: the lowest stock-to-sales ratio (0.13), 14 of 26 critical stockout-risk SKUs, the highest Sephora rating (4.31), and a 10% tariff premium.
4. **Consumer review velocity is a validated leading demand signal,** at least within its skincare-only scope. Review volume grew 6.5x from 2015 to 2020, tracking and preceding the same period's demand pressure identified independently in the operational stockout data.
5. **Near-shoring can reduce both cost and lead time simultaneously** in this portfolio's transportation data. Sea routes are both the cheapest and fastest mode observed, undercutting the usual assumption that near-shoring necessarily carries a cost premium.

Recommendations

- Initiate dual-sourcing qualification for Supplier 4's SKU portfolio as the top operational priority, independent of any tariff-mitigation timeline.
- Recalibrate safety stock and reorder points for skincare SKUs first, using the Reorder Point Table's stock-to-sales ranking as the starting priority list.
- Integrate Sephora-style review-velocity monitoring into upstream demand-planning tools (Anaplan or o9 Solutions) as a leading indicator, understanding that this signal is currently

validated for skincare only and would need equivalent review data for cosmetics and haircare before being extended to those categories.

- Treat quality-risk audits (defect rate, inspection failure) and tariff-exposure reviews as two separate, parallel workstreams rather than a single combined risk register, given Finding 2 above.

9. Future Work

- Extend the operational dataset beyond 100 SKUs to support statistically generalizable (rather than purely directional) conclusions, and to enable a genuine predictive model, for example a stockout-event classifier, once sample size supports it.
- Acquire or construct a review dataset with real coverage of cosmetics and haircare, to test whether the review-velocity demand signal validated for skincare in this project generalizes to other categories.
- Wire the already-built Tariff Sensitivity parameter into the dashboard as a genuine what-if layer alongside the fixed `tariff_impact_coefficient`, letting users compare the policy-grounded baseline against their own hypothetical scenarios rather than choosing one over the other.
- Recalibrate the `tariff_impact_coefficient` model against a real enterprise's actual margin structure (70–80% baseline, not this sample's ~98%) to convert the current directional ranking into an enterprise-usable dollar-impact forecast.
- Revisit the four remaining scope decisions listed in Section 7 (the Tariff Impact Coefficient filter, waterfall bookend bars, `category_bridge.csv`, and margin recovery velocity) as candidates for a future iteration, now that they are documented rather than simply absent.

10. References & Acknowledgements

Data Sources

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Generative AI Disclosure

Claude (Anthropic) was used throughout this project as a development, QA, and documentation partner, consistent with the disclosure practice established in every prior deliverable. Specific uses included: brainstorming how to align raw Kaggle variables with enterprise ERP/WMS/S&OP terminology during scoping; executing Python profiling scripts directly against the uploaded datasets to produce real null-count and distribution statistics (every numerical finding in this report is derived from that actual data execution, not an AI-generated estimate); stress-testing the logic of the category-based tariff proxy strategy; identifying and helping resolve the Tableau filter-routing and legend-formatting issues described in Section 7; and reviewing dashboard and datafolio drafts for consistency, clarity, and adherence to the project's brand and disclosure standards. All architectural decisions, domain interpretations, and stakeholder-alignment choices reflect the author's own analytical judgment and 20+ years of professional ERP and supply chain consulting experience; Claude's outputs were used as a development aid and were reviewed and verified before inclusion, in line with the responsible and transparent use of generative AI described in this program's supportive-tools guidance.